HVAC Trainer Kit



Category: **Electrical & Electronic Components & Kits** Type: **Full SOP** Priority: **High**

1 labs

Covers these lab items: HVAC Demonstration Kit

What it is used for

To demonstrate the basic components and operating principles of heating, ventilation, and air conditioning systems using an HVAC trainer kit. To perform supervised measurements and practice safe troubleshooting on the training setup.

Main parts

- Training panel and frame
- Refrigeration compressor and circuit, if fitted
- Condenser and evaporator sections
- Fans, ducts, and air filters
- Thermostat, sensors, and control switches
- Pressure gauges, temperature meters, and electrical terminals

Before you start

1. Read the trainer manual, circuit diagram, and laboratory instructions.
2. Inspect the power cord, plug, earthing, terminals, guards, and emergency stop if fitted.
3. Check that refrigeration pipes, hoses, and fittings show no visible damage or leakage.
4. Ensure the trainer is placed in a ventilated area with sufficient clearance around air inlets and outlets.
5. Confirm that the refrigerant circuit is sealed and that only authorized personnel service it.
6. Prepare suitable measuring instruments and check their condition before use.

How to use

1. Keep the trainer switched off while checking connections and components.
2. Identify the compressor, condenser, evaporator, fans, sensors, and control devices using the diagram.
3. Verify that electrical connections and protective earthing are correct before energizing.
4. Set the trainer controls to the demonstration settings specified by the instructor or manual.
5. Switch on the main supply and start the system using the approved procedure.
6. Observe the compressor, fans, airflow, indicators, and temperature readings during operation.
7. Allow the system to stabilize before recording inlet and outlet temperatures or other specified readings.
8. Measure voltage, current, pressure, or airflow only with suitable instruments and approved test points.

9. Compare readings with the expected operating behavior and discuss any differences with the instructor.
10. For troubleshooting exercises, switch off and isolate power before inspecting electrical components.
11. Do not adjust refrigerant charge or open the sealed refrigeration circuit during routine training.
12. Switch off the trainer, isolate the supply, and allow moving or hot components to stop and cool.
13. Record results, return controls to their safe positions, and tidy the training area.

⚠ Safety

- Only trained learners under instructor supervision should operate the trainer.
- Use correct earthing and never touch exposed live terminals.
- Isolate power before changing wiring or handling electrical components.
- Fans and compressor components may start automatically; keep hands and tools clear.
- Do not loosen refrigerant fittings or release refrigerant; servicing must be done by authorized personnel.
- Avoid contact with hot pipes and cold refrigerant lines, which may cause burns or frost injury.
- Stop immediately if there is smoke, an electrical smell, unusual vibration, or suspected refrigerant leakage.

Care & storage

- Switch off and isolate the trainer after each session.
- Clean filters, vents, and external surfaces as instructed.
- Inspect cables, terminals, hoses, and guards for wear or damage.
- Keep instruments and accessories in their designated storage places.
- Arrange scheduled maintenance and calibration for gauges and sensors.

Common mistakes to avoid

- Starting the trainer before checking earthing and connections.
- Blocking air inlets or outlets during operation.
- Touching live terminals or moving fan parts during measurements.
- Using unsuitable meters or incorrect measurement points.
- Opening the refrigerant circuit or changing charge without authorization.
- Ignoring unusual readings, noises, or signs of leakage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.